

**Only 1- and 2-loops** =  $\frac{1}{3} \times \begin{pmatrix} 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 6 \end{pmatrix},$

**1- and 2-loops and a 1-cycle and a 2-cycle in the other direction =**

**Everything except 0-loops** =  $\frac{1}{3} \times \begin{pmatrix} 6 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 9 \\ 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 \\ 9 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 \\ 9 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 6 \end{pmatrix},$

$$\text{All } 1\text{'s} = \frac{1}{3} \times \begin{pmatrix} 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & 3\omega^2 & 0 & 0 & 0 & 3\omega^2 & 3\omega^2 & 0 & 0 \\ 0 & 0 & 3\omega & 3\omega & 0 & 0 & 0 & 3\omega & 0 \\ 0 & 0 & 3\omega & 3\omega & 0 & 0 & 0 & 3\omega & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & 3\omega^2 & 0 & 0 & 0 & 3\omega^2 & 3\omega^2 & 0 & 0 \\ 0 & 3\omega^2 & 0 & 0 & 0 & 3\omega^2 & 3\omega^2 & 0 & 0 \\ 0 & 0 & 3\omega & 3\omega & 0 & 0 & 0 & 3\omega & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \end{pmatrix}$$